

SYSTEMS & AUTOMATION PORTFOLIO · 2026

PORTFOLIO PROJECT · AI & AUTOMATION

AI Voice Reservation Concierge

A voice-driven reservation system that answers the phone as a calm, butler-style host, takes a restaurant booking from greeting to confirmation entirely hands-free, checks live table availability across the floor, and manages cancellations, changes, and look-ups — writing every reservation to a live spreadsheet with no human picking up the line.

WHAT IT DOES

Automates time-aware call greeting, natural-language booking capture, live table-availability matching, reservation-ID generation, and spreadsheet sync for cancellations, updates, and look-ups.

TOOLS USED

VAPI · ElevenLabs · n8n · Webhooks & APIs · Google Sheets · JavaScript · Deepgram (speech-to-text)

WHO BENEFITS

Restaurants, clinics, salons, and any small business that loses bookings to missed calls and wants a 24/7 voice agent without hiring more front-desk staff.

KEY FEATURES

Time-of-day greeting · Live 15-table availability with overflow · Auto reservation IDs · Phone-based retrieval · Confirm-once dialogue · Four-action coverage (book / cancel / update / look-up)

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What Is This Project?

Every missed phone call at a busy restaurant is a lost booking. The front desk is mid-service, the line rings out, and the guest moves on to the next place. The manual chain — answer, take the name, check the book, find a table, confirm, write it down — only works when someone is free to pick up.

This system replaces that entire chain with a voice agent named Jarvis. The moment a call comes in, VAPI greets the guest by the time of day in a warm, butler-style voice, listens, and understands what they need. Behind the voice, an n8n workflow with four dedicated webhooks does the real work: it checks live table availability, books the table, generates a tidy reservation number, and writes everything to a Google Sheet — or, just as easily, cancels, changes, or looks up an existing reservation by phone number. No human ever has to pick up the phone.

4 Reservation Actions Book · Cancel · Update · Look-up	15 Tables Tracked 6 small + 9 large, per time slot	0 Humans on the Line Fully hands-free intake	24/7 Availability Never misses a call
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Why this build: *small businesses lose real revenue to unanswered phones, yet a full reservation platform is overkill and a human receptionist is costly. This system was designed around how a real venue actually takes bookings — including the awkward parts most demos skip, like a guest who misspells their name, a fully-booked slot, or someone calling back to change a time.*

The 4 Phases — System Overview

The system runs as four connected capabilities, each a self-contained webhook flow:

PHASE 1

Call Intake & Time-Aware Greeting

VAPI answers the call, calculates the local time, and greets the guest with the right salutation — morning, afternoon, or evening — in a calm gentleman's voice, then identifies which of the four actions the guest needs.

PHASE 2

Booking & Live Availability

For a new reservation, the booking webhook reads the live sheet, counts how many of the 6 small and 9 large tables are taken for that slot, assigns the right table (small parties overflow to large when needed), generates a reservation ID, and writes the confirmed booking.

PHASE 3

Retrieval & Changes

Cancellations, updates, and look-ups find the guest's reservation by phone number — tolerant of formatting like leading zeros and country codes — then read the name and details back so the guest can confirm before any change is finalised.

PHASE 4

Confirmation & Source of Truth

Every action returns a spoken confirmation in VAPI's required format and writes the result to a single Google Sheet — the live source of truth the whole venue can watch update in real time.

How It Works — Step by Step

PHASE 1 · CALL INTAKE & GREETING

From Ring to Understanding

1

The call connects and Jarvis greets by time of day

VAPI answers instantly, derives the local hour, and opens with “Good morning / afternoon / evening — thank you for calling The Gilded Fork. This is Jarvis, at your service.” — in a warm ElevenLabs butler voice.

2

Speech is transcribed and the intent is identified

Deepgram converts the guest's speech to text in real time; the assistant decides whether they want to book, cancel, change, or check a reservation, and calls the matching tool.

3

Details are gathered — and confirmed once

Jarvis asks one gentle question at a time, confirms the name spelling and phone number a single time (no robotic repetition), and speaks dates naturally — “Friday, the twenty-sixth of June” rather than “2026-06-26.”

PHASE 2 · BOOKING & AVAILABILITY

A Table Is Never Double-Booked

4

Live availability is checked against the sheet

The booking webhook reads every confirmed reservation for that date and time and counts how many of the 6 small and 9 large tables are already taken — never a cached guess.

5

The right table is assigned, with overflow logic

Parties of 1–2 take a small table; 3–8 take a large one. If all small tables are full, a couple overflows into a large table. Only when both types are full does Jarvis offer another time.

6

A reservation ID is generated and the booking is saved

Each confirmed booking gets a compact ID — e.g. U2626200021 (month, day, year, time, party size, weekly count) — which Jarvis reads back to the guest, and the row is written to Google Sheets.

How It Works — Retrieval & Confirmation

PHASE 3 · RETRIEVAL & CHANGES

Find by Phone, Confirm by Voice

7

The guest is found by phone number

For a cancellation, update, or look-up, Jarvis asks for the phone number first and searches the sheet — matching tolerantly so leading zeros and +966-style country codes still resolve to the right booking.

8

The reservation is read back for confirmation

The system returns the guest's name, reservation ID, date, time, and party size, and Jarvis reads them back so the guest can confirm it is the correct booking — no need to memorise the reservation number.

9

The change is applied safely

Updates change only the fields the guest mentions, keeping the rest intact; cancellations mark the row cancelled rather than deleting it, preserving a clean record.

PHASE 4 · CONFIRMATION & SOURCE OF TRUTH

Spoken Back, Written Down

10

A spoken confirmation is returned to the guest

Each webhook replies in VAPI's required result format, echoing the call's tool-call ID so the assistant speaks the outcome back gracefully and in the butler's tone.

11

Google Sheets stays the single source of truth

Every booking, change, and cancellation lands in one sheet — visible, exportable, and easy to demo live as rows appear during a call.

12

Empty and stray calls are filtered out

The booking flow ignores VAPI status events and refuses incomplete submissions, so the record never fills with blank or junk reservations.

The Tools Working Together

Seven tools are connected across this system, each with a specific, non-overlapping role:

V

VAPI — The voice layer

Answers the call, runs the conversation, calls the right tool, and speaks every response in the butler persona.

E

ElevenLabs — The gentleman voice

Provides the calm, mature English-butler voice that gives Jarvis his character on every call.

D

Deepgram — Speech-to-text

Transcribes the guest's speech in real time so the assistant can understand names, numbers, dates, and intent.

n

n8n — The automation engine

Hosts the four webhook flows, runs the availability and matching logic, and orchestrates every action end to end.

W

Webhooks & APIs — The integration layer

Each VAPI tool calls its matching n8n webhook in real time; responses return in VAPI's required tool-result format.

G

Google Sheets — The master registry

Stores every reservation with status and assigned table type — the live source of truth for availability and retrieval.

J

JavaScript — The logic engine

Code nodes parse the VAPI payload, run the table-availability and overflow logic, build reservation IDs, and match by phone.

Who Benefits from This System?

Built with three groups in mind — each gets something different out of it:

Restaurant Owners A phone line that never goes unanswered. Bookings are captured 24/7, even mid-service or after hours, with every reservation logged automatically and visible at a glance.	Front-of-House Staff No more juggling the phone during a rush. Routine bookings, changes, and cancellations are handled by Jarvis, freeing the team to focus on the guests already in the room.	Operations & Owners-at-Scale A template that drops into any booking business — clinics, salons, tours — with the same four actions and a live spreadsheet, at a fraction of the cost of a reservations platform or extra staff.
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Why This Matters

- ✓ **No booking is ever lost to a missed call** — the line is answered 24/7, instantly and consistently.
- ✓ **The voice does the work, not the staff** — greeting, capture, and confirmation happen with no one on the line.
- ✓ **A table is never double-booked** — live availability is checked against the sheet before every confirmation.
- ✓ **Guests are found even without their reservation number** — phone-based, format-tolerant retrieval.
- ✓ **Everything stays in one live spreadsheet** — visible, exportable, and updated in real time.
- ✓ **The persona stays consistent** — a calm, gracious butler on every single call.

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